# AliveCor SDK Overview

**9/10/2025**

**Intended Use**

The Kardia mobile SDK lets someone who wants to build a mobile app integrate the *Kardia* app software building blocks to access our devices, initiate ECG recordings, display the waveform, interpret the results using AliveCor's AI algorithms, and display the results to the user. The SDK can perform the entire turnkey recording flow all at once with one library call. The SDK package is available for Android and iOS.

**Functional overview**

The structure of the AliveCor SDK is built around the recording experience of our *Kardia* consumer app. We've packaged the entire recording experience including Bluetooth pairing, permissions setting, error handling, user display during the recording, and generating the AI determination (i.e. normal, possible afib, bradycardia, tachycardia) into a single library call. This allows the mobile app builder to build the ECG recording into their desired menus/workflows to meet their specific needs, without the need to manage any of the low-level details or manage any error cases such as Bluetooth errors, interrupted recording, etc. This approach greatly minimizes the development effort required.

When the ECG recording is initiated, the AliveCor SDK takes over the UI of the mobile device and performs the entire recording sequence. Once the recording is complete, control of the UI is returned to the partner app. At the end of the recording, a raw data file is created on the mobile device that the mobile app can manage. There are also options to display rhythm strip views to the user, and to create PDFs of the recording. These raw data files and PDFs can then be sent to the partner's backend system via the partner's mobile app. The partner's app is responsible for managing data transfer from the mobile device to the partner's backend system, and/or setting up a database on the mobile device to store the data locally. AliveCor does not store or process any patient data in our servers, it is completely contained on the mobile device and is under the control of the partner's app.

Mobile apps built with our SDK are required to authenticate periodically with AliveCor on a per subscriber basis. The details of this process and the requirements on the part of the partner are described later in this document. See **SDK Authentication using the Kardia SDK Auth Server.**

**Sample Apps**

The AliveCor SDK packages are provided with sample apps to show examples of how to authenticate the SDK and use the SDK functions (authentication, ECG recording, PDF generation, etc.). Please note that the sample apps we provide may contain imbedded URLs and SDK credentials that are unique to AliveCor's test environment, so when cutting and pasting sample app code into your apps, be sure to replace those URLs and credentials with ones that point to your authentication server (see SDK Authentication section for details).

**Regulatory Considerations**

The AliveCor SDK is cleared for use in the following counties and regions:

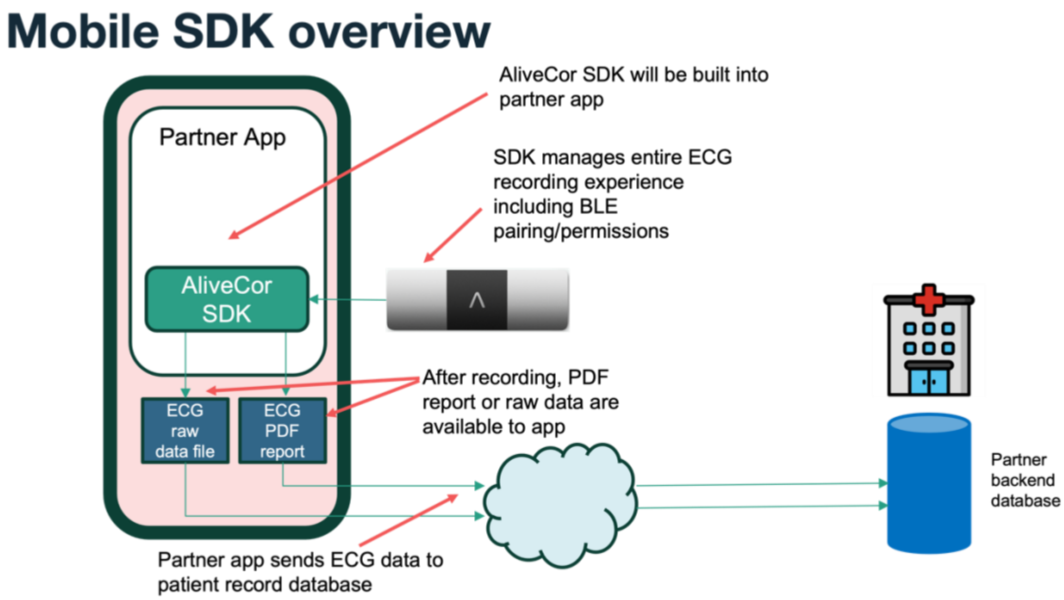
- United States
- European Union countries
- Switzerland
- United Kingdom
- India
- KSA
- Thailand
- Taiwan
- South Africa

Use in approved markets is limited to the prepackaged recording flow, without customizations. Your company will need to evaluate any regulatory requirements for apps that are built with our SDK, depending on the intended country of use.

**Feature Matrix**

Here is a high level overview of the features provided by the AliveCor SDK packages. This is bundled as "Kardia Kit Lite" in the SDK package. Kardia Kit is no longer supported.

| Feature | What the user sees in the UI | Data the mobile app can access |
|---|---|---|
| Turnkey recording flow | Consumer app recording experience from pairing through recording completion, including real time ECG waveform, BPM, AI determination view | Raw ECG time series data, AI determination, and BPM (available post recording) |
| Rhythm strip view with AI determination (optional-controlled by partner app) | Preview snapshot or fully scrollable rhythm strip view including AI determination and BPM (available post recording) | Raw ECG time series data, AI determination, and BPM (available post recording) |
| Generate PDF report (optional-controlled by partner app) | Controlled by partner app. Can be visible or invisible to user (available post recording). | PDF file of the completed ECG (available post recording) |




**AliveCor ECG device compatibility:**
AliveCor's SDK is compatible with the following AliveCor devices:

KardiaMobile
KardiaMobile 6L
KardiaMobile Card

Allowed device type(s) is an SDK configuration parameter that AliveCor sets as part of the SDK authentication credentials. The partner needs to identify which devices the partner app wants to support when the authentication credentials are set up. The default is KardiaMobile 6L only.

Note that not all ECG devices are available in all markets, so device availability and SDK clearance both need to be considered on a county by country basis.

**AI Arrhythmia detection**
AliveCor's SDK includes the following arrhythmia detections, which run on the mobile device:

- Atrial Fibrillation
- Bradycardia
- Tachycardia
- Normal Sinus Rhythm
- Sinus Rhythm with Premature Ventricular Contractions (PVCs)*
- Sinus Rhythm with Supraventricular Ectopy (SVE)*
- Sinus Rhythm with Wide QRS*

* These additional arrhythmias are disabled by default unless specifically requested by the partner. In the technical documentation, the first 4 as a group are referred to as AIv1. The entire list of 7 are referred to as AIv2 in the technical documents. If you want to use AIv2 (full list of all 7 arrhythmias results), please indicate this in your request for credentials.

**iOS and Android Device Compatibility**

The Kardia SDK for iOS and Android is built on our consumer facing *Kardia* app. The devices we test with and guarantee SDK compatibility with are listed here. AliveCor cannot guarantee compatibility with devices outside of our tested device list.

**Staging and production environments**

AliveCor provides a staging or "sandbox" environment to facilitate SDK app development and testing, separate from our production environment. No live patients are allowed on staging. Access to the SDK in a staging environment can be provided upon completion of a test agreement. Moving to production requires a commercial agreement to be in effect and requires AliveCor to approve a partner provided test report for regulatory purposes. Instructions for the technical modifications needed to move to production will be included in your Box folder.

**SDK Support**

AliveCor has created a developer support program for requesting SDK support. In addition to entering support tickets, developers can also search for help articles and find other resources. Please send the emails of the software developers who will need access to this portal to your AliveCor representative so they can be added to the developer's support program. Once invited and your account is setup, support tickets can be entered at https://alivecor.atlassian.net/servicedesk/customer/portal/4

**SDK Authentication using the Kardia SDK Auth Server:**

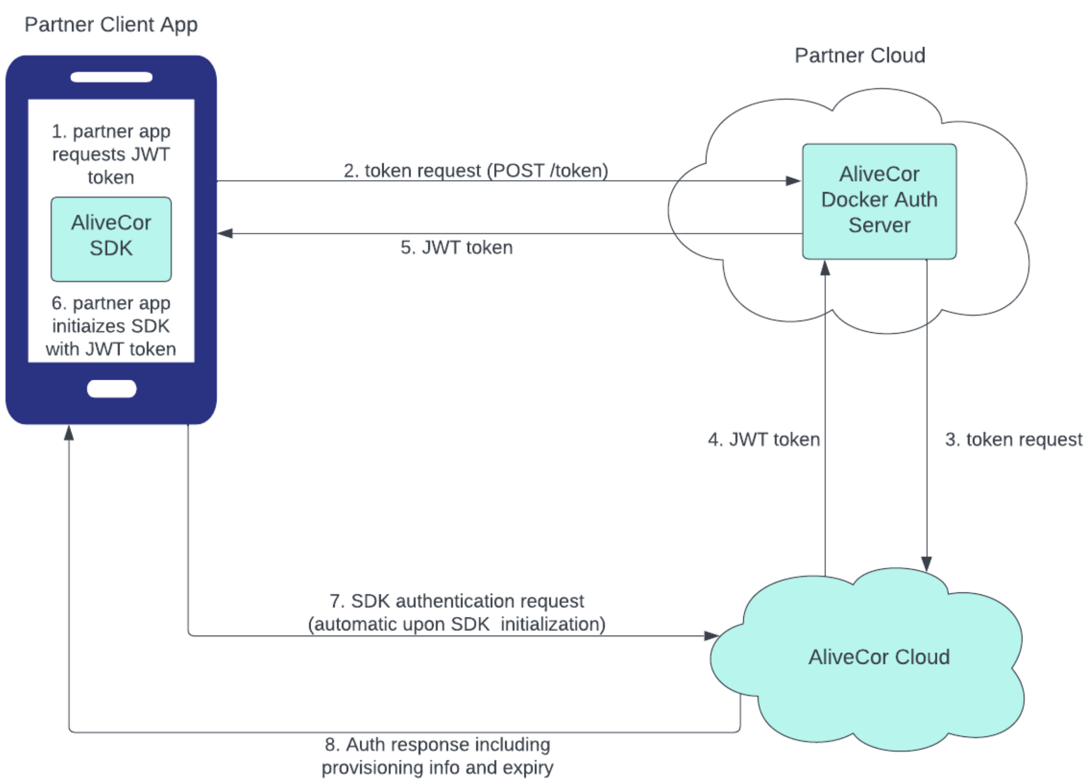
Upon providing App ID/ BundleID, to AliveCor, we will provide the following credentials which will be required to build the Authentication Server and generate tokens (see "Setting up the Authentication Server" below for details).

KARDIA_ACCESS_KEY
KARDIA_SECRET_KEY
partnerId
teamId

These credentials will be provided in your SDK delivery folder. When submitting the request for credentials, please provide the following information:

1. Bundle id(s) for your apps
2. Allowed end user ECG device types (KardiaMobile, KardiaMobile 6L, KardiaMobile Card)
3. AIv1 or AIv2 algorithm
4. Staging or Production environment

## Authentication sequence




In order for Kardia SDK partner applications to authenticate with the Kardia Backend, JWT tokens are required.

In order to fetch a JWT token, the partner is required to set up an authentication server, in their own cloud environment, that will receive token generation requests from your app. The self-hosted auth server will then communicate with the AliveCor Cloud to authenticate the token request and return a token to the requesting client.

When the SDK is initialized with the JWT token, the SDK will then automatically “phone home” to AliveCor’s backend to gather its provisioning information including which ECG devices the SDK is authorized to communicate with, which AI determination result set is allowed, and the authentication expiration date of 7 days.  The 7-day expiration is designed to allow for offline operation of the SDK for up to a week.

**Setting up the Authentication Server:**
The partner will need to pull an AliveCor created image and run it on the cloud provider of your choice.

3 Environment Variables are required to run the server:

KARDIA_ACCESS_KEY
KARDIA_SECRET_KEY
KARDIA_ENV (“sandbox” or “production”)

Optional Environment Variable: PORT (server will default to Port 8080, if changed update the docker run command)

Here are the docker commands to load and run the server:

1. docker pull kardiasdk/kardia-auth-server
2. docker run --env KARDIA_ACCESS_KEY=$KARDIA_ACCESS_KEY --env KARDIA_SECRET_KEY=$KARDIA_SECRET_KEY --env KARDIA_ENV=$KARDIA_ENV -p 8080:8080 kardiasdk/kardia-auth-server ./kardia-auth-server

This sets up a /token API running on the port above that can be used as an external endpoint that can be interacted with from your client application to request JWT tokens. At this point ensure your Kardia SDK Application is pointed to your new server, and token requests should work as expected. **Note:** if you are copying our sample apps they contain authentication server URLs specific to the AliveCor lab environment and they will need to be replaced with your authentication server URL.

**/token Endpoint details :**

Note that “patientMrn” can be any alphanumeric value up to 32 characters. You can re-use the same value for each token request. Here is an example token request/response example:

POST body will contain 4 values,
{
  "bundleId": "bid1",
  "partnerId": "examplePartnerId",
  "patientMrn": "ExamplePatient MRN",
  "teamId": "exampleTeamId"
}
Response
{
  "jwt":
    "eyJhbGciOiJIUzUxMiJ9.eyJidW5kbGVJRCImJpZDEiLCJleHAiOjE2MDMxNjU4NjksImlhdCI6MTYwMzE2NDY2OSwianRpIjoiMjA2N2M2MmItZDY1MS00MjZlLWE4MjgtMWViMjc4YmE3ZTUxIiwibXJuIjoiSmFycmVkIiwibmJmIjoxNjAzMTY0NjY5LCJ0ZWFtSUQiOiJGczBaSzFyYVppNUtpZE50Q3lLOWI0bnloNmdrcTRhaCJ9.nqIN9_QaaU4yOMpad68yP2v6qtcep1P5E_4m0UaHy7X5tp6GsR9Pq6z4lsdGmWHea4ao99EEf3Vll0Y5w4E4yQ"
}

This JWT token is then used to initialize the SDK which triggers an automatic authentication sequence to AliveCor's servers.